

FALCON's Response to Festo for Supply Of Conveyor Automation – 2K Option - 3

August 1, 2024

Doc. No. - Sales/FR/PT/007/R0



Kind Attention –

M/s Festo

Offer Ref: F24-00322 Option 3; Date: 01-08-2024

Subject – Techno -Commercial Offer of Conveyor Automation System 2K PPH

We are pleased to submit our Techno-Commercial Offer in response to your RFP.

As you will note, we have done an in-depth data analysis and evaluated various solution options best suited for your requirements, along with the information collected during the meetings and discussions with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to being your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Intra-logistics Automation Technologies and references.

To conclude, I would like to add my personal commitment on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

Best Regards,

Sandeep Bansal

Chief Business Officer

**Response to RFP for
Conveyor Automation 2K PPH Throughput**

Proposal Reference – F24-00322 – Option 3

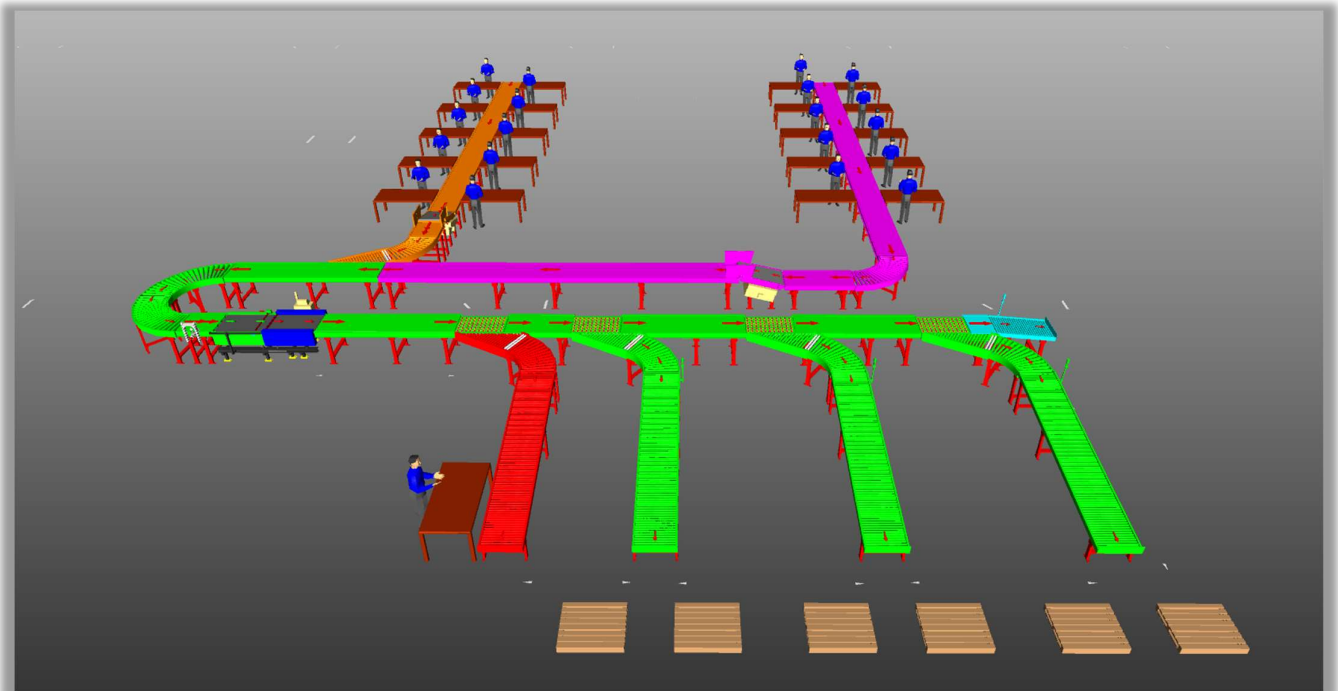


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1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour
3	AC	Alternating Current
4	DC	Direct Current
5	PLC	Programmable Logic Controller
6	IT	Information Technology
7	BOQ	Bill Of Quantity
8	I/O	Input/ Output
9	PDP	Power Distribution Panel
10	PC	Personal Computer
11	UPS	Uninterrupted Power Supply
12	SWEDI	Swivel Wheel Divert

2. Executive Summary

Falcon Autotech is pleased to confirm its great interest in responding to this Festo RFQ of Conveyor Automation System. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening, understanding your needs, and ensuring this project's success.

Following the same objective for the Festo Conveyor automation system, we are happy to offer a compliant solution meeting all technical and operational requirements at competitive price, delivering key results.

Our solution is based on the following key characteristics:

Conveyor Automation System 2K PPH:-

- *Two Dedicated lines for loading final order carton (1st presently & 2nd in future).*
- *Main Highway line Is integrated with Weighing system.*
- *SWEDI Diverters are placed to Divert the carton at final destinations.*

3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0



- 5** Product Lines
- 15+** Countries with Live Installations
- 1800+** Total Installed Systems Globally
- 1Million+** Sorts Per Hour
- 600+** Employees

Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

Long-standing Relationships with industry leaders



















News & Articles



FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.



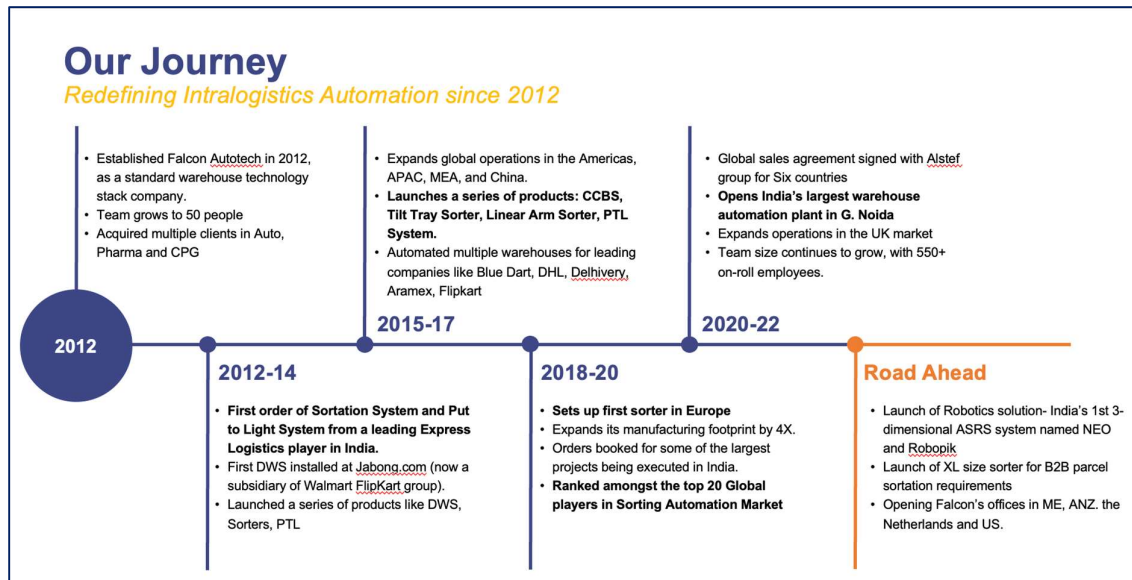
FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for Festos and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its Festos through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our Festos, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



Product and Solutions

With 100% focused on Parcels, Eaches, Totes, Bags, Cartons



SORTATION SOLUTIONS

- Cross Belt Sorter
- Linear Arm Sorter
- Swivel Divert Sorter
- Tilt Tray Sorter
- Popup Sorter
- Sweep Sorter
- Pusher Sorter



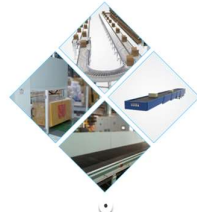
PICK/PUT TO LIGHT SYSTEMS

- PTL Module
- Racks
- Conveyors
- Hand Scanners
- Printers
- Peripheral Displays



DIMENSIONS & WEIGHT SCANNING SYSTEMS

- Cubizon Series
- R, R-Eco, R-Thru, R-Cross
- Dynamic Profiles
- Mini (600MM)
- Jumbo (1200MM)



CONVEYOR SOLUTIONS

- Belt Conveyors
- Roller Conveyors
- Modular Conveyors
- Special Application Solutions



ROBOTICS

- NEO
- Robopick

Powered by

FACtS

Sort IT

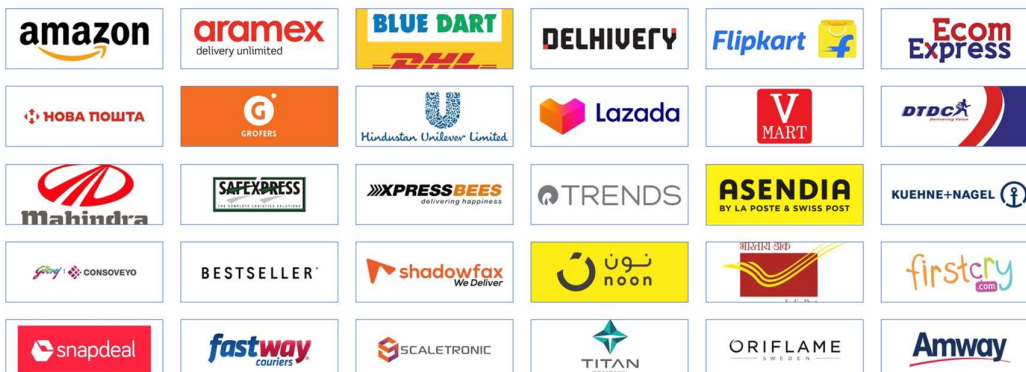
Control IT

Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

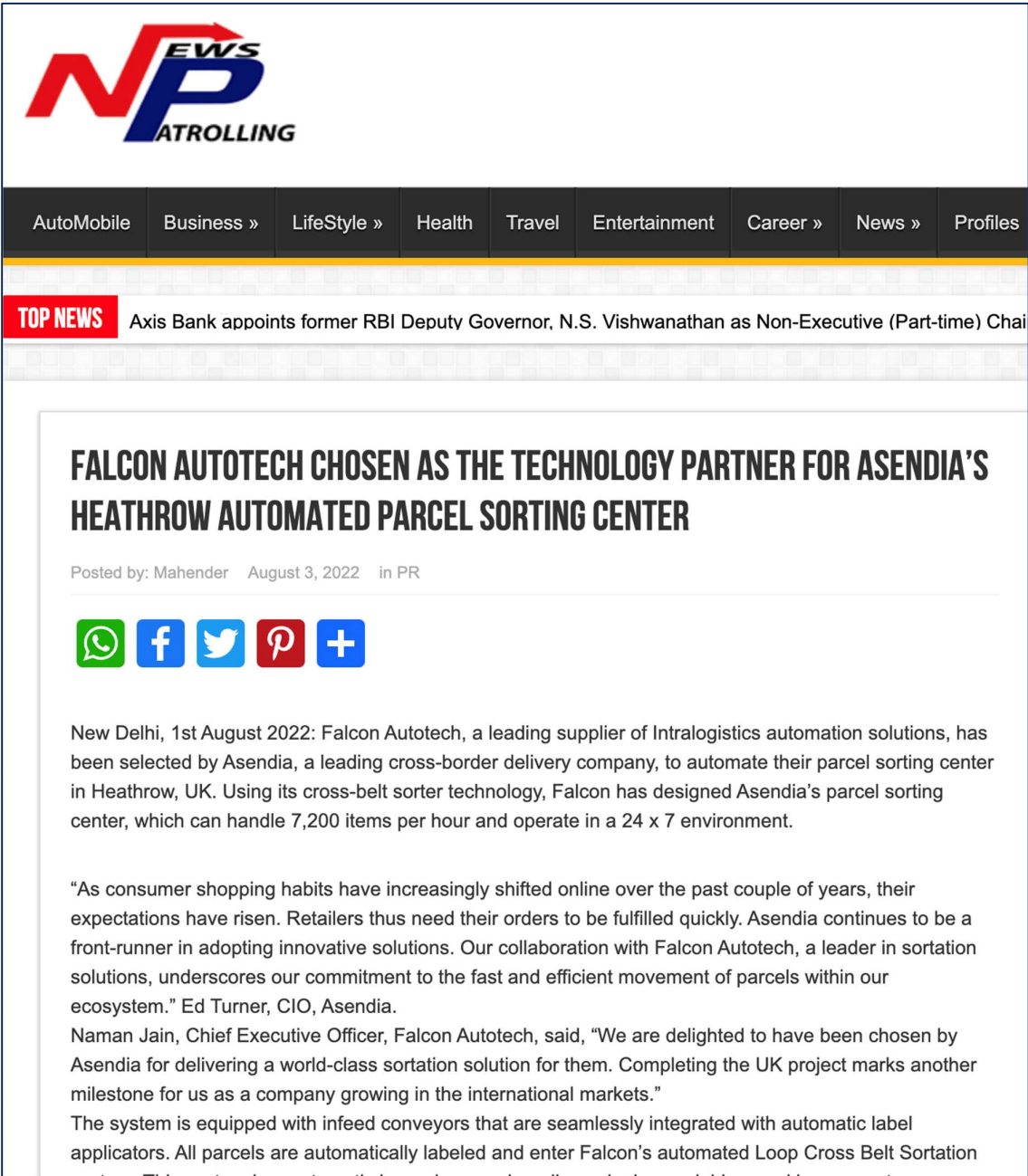


With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.



4. Falcon’s Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of **the World’s largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focuses on Sortation Solution –

5.1 Project 1- (CEP Festo, India)

The System is equipped with two fully automated and interconnected Sub-systems. Sub-System 1 is designed for handling large B2B boxes and E-commerce shipment bags while the Sub-System 2 is designed to handle Small E-commerce packages.

Solution Specifications –

- 48,000 PPH (Double Deck CBS- Shipment Sorter)
- 17,000 PPH (Double Deck CBS- Bag Sorter)
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- | | |
|---|--------------------------------------|
| • 2 Sets of Double Decker CBS Sorters | • Modular Belt Conveyors |
| • Mezzanine Structures (one large mezz on to install a double decker Sorter). | • Spiral Chutes with Braking rollers |
| • Automated Singulators | • 5-Sided Scanning Tunnels |
| • Fully Automatic Inductions | • High speed weighing conveyors |
| • Semi-Automatic Inductions | • Direct Bagging Chutes |
| • Telescopic belt conveyors | • Put to Light Chutes |
| • PVC Belt Conveyors | • Volume Distribution systems |
| | • High Availability Server Systems |
| | • WCS |



5.2 Project 2- (Festo – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

In 2019, Festo was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

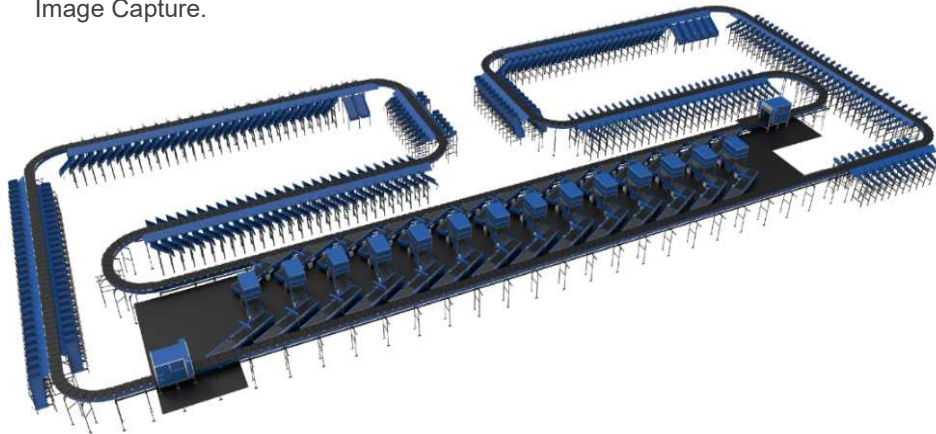
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System



5.3 Project 3- (Ecommerce Festo, India)

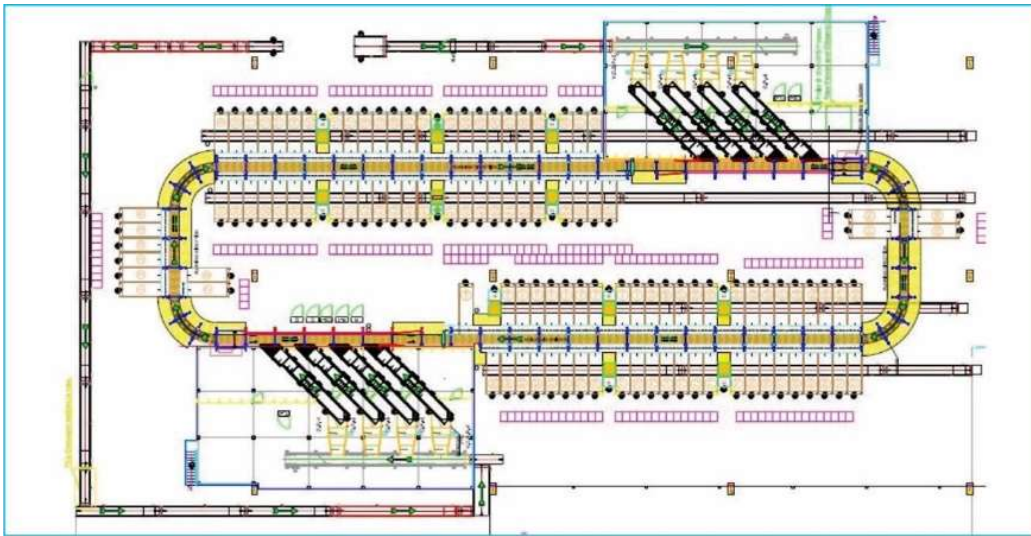
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



5.4 Project 4- (CEP Festo, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with an automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.5 Project 5- (CEP Festo, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

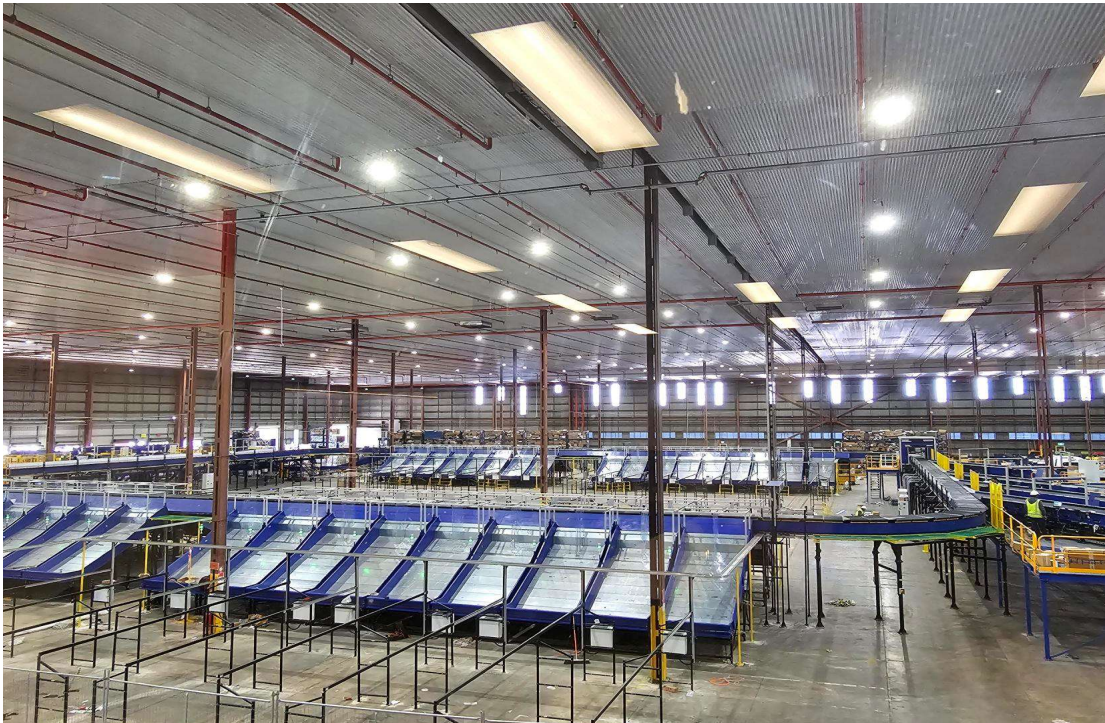
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



6. About Falcon

THE COMPANY

ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

OUR CORE COMPETENCIES

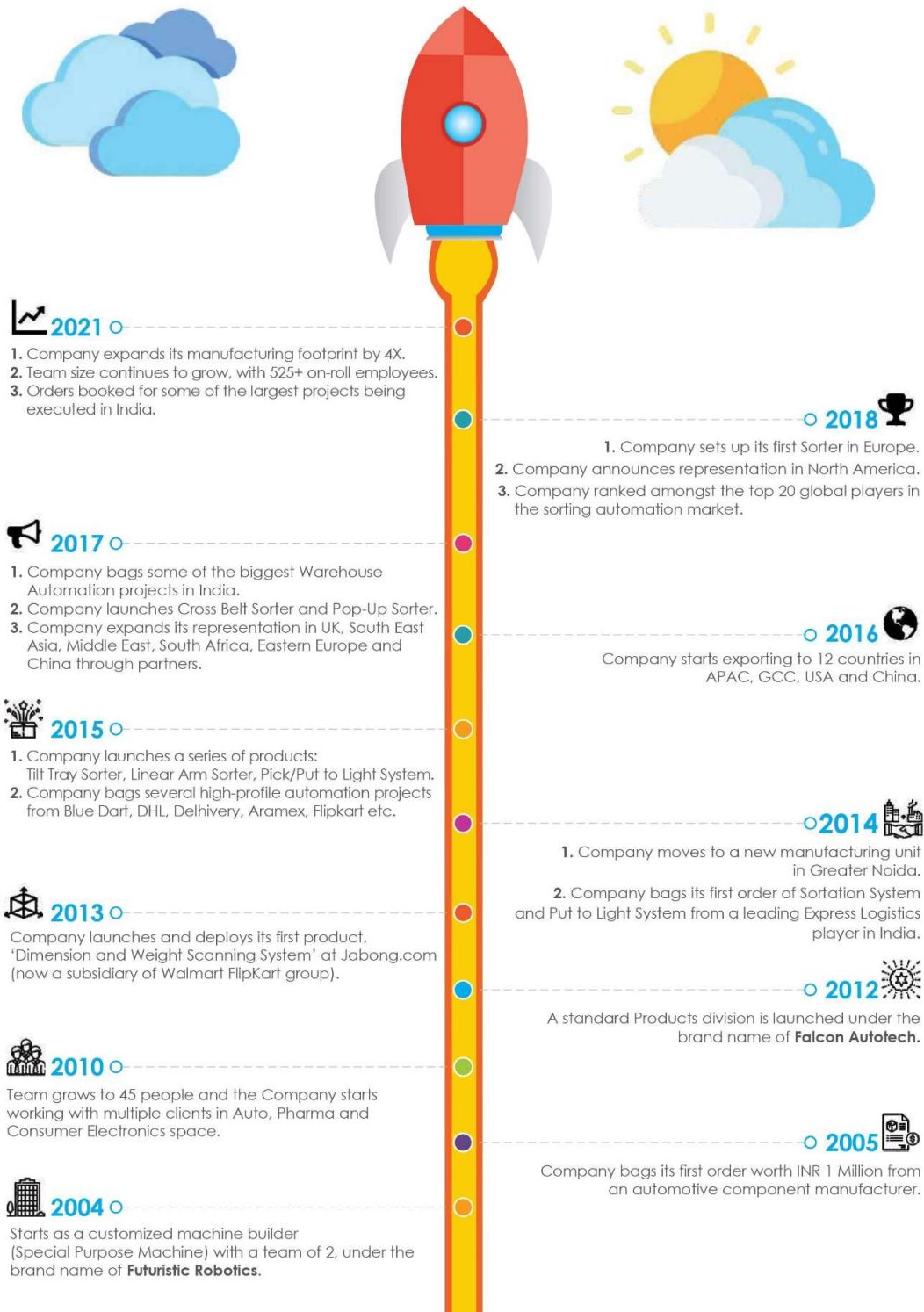
Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more. At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability to harness our true potential generated by culmination of Supply Chain Know-How, quality Hardware and flexible Software under one roof.



COMPANY HISTORY



KEY STATISTICS



04

Product Lines



525+

On-Roll Employees



290+

Engineering Team



400000+ Sq Ft

Manufacturing Space



20+

Tier A Clients



15+

Countries with Live Installations



1800+

Total Installed Systems Globally



35

Representations in 4 Continents and 35 Countries



1 Million+

Sorts Per Hour of combined Sorting Capacity
installed Globally

OUR PRODUCT PORTFOLIO

01

SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

DIMENSION & WEIGHT SCANNING SYSTEMS

STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

DYNAMIC CONVEYOR BASED SYSTEMS

OUR 8 MODELS



OUR PRODUCT PORTFOLIO

04

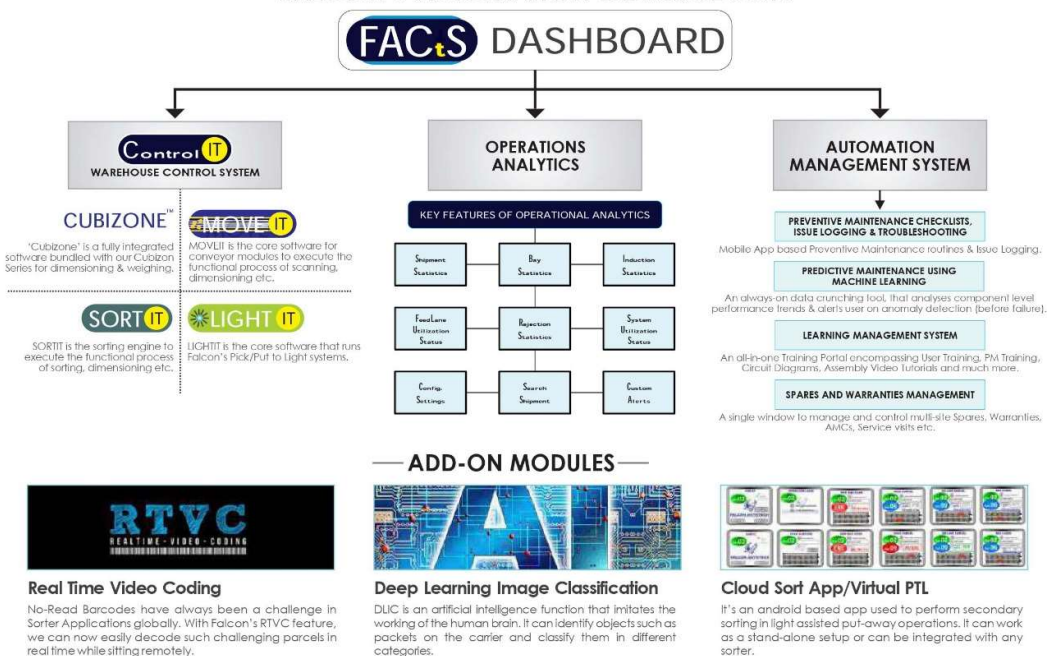
CONVEYOR AUTOMATION SOLUTIONS

BELT CONVEYORS	ROLLER CONVEYORS	SPECIAL APPLICATION SOLUTIONS
❖ PVC BELT ◆ Straight Conveyor ◆ Spacing Conveyor ◆ Accumulation Conveyor ◆ Strip Belt Merge- 30/45/60 ◆ Solid Belt Merge- 45 ◆ Incline Belt Conveyor ❖ MODULAR BELT ◆ Straight Conveyor ◆ 90 Degree Belt Curve ◆ 180 Degree Belt Curve	◆ Straight Conveyor ◆ Gravity Roller Conveyor ◆ MDR Accumulation ◆ Angled Roller Conveyor ◆ Curved Conveyor ◆ Spur Merge Conveyor ◆ Swivel Wheel Diverter ◆ Pop Up Belt Diverter ◆ Pusher Unit	◆ Aligning Conveyor ◆ One to Two Split Conveyor ◆ Two to One Merge Module ◆ Bulk Distribution-ARB ◆ Bulk Distribution-Telescopic ◆ Bulk Distribution-Electric Arm ◆ Telescopic Belt Conveyor
		

05

SOFTWARE SUITE

FALCON'S AUTOMATION CONTROL SUITE



INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



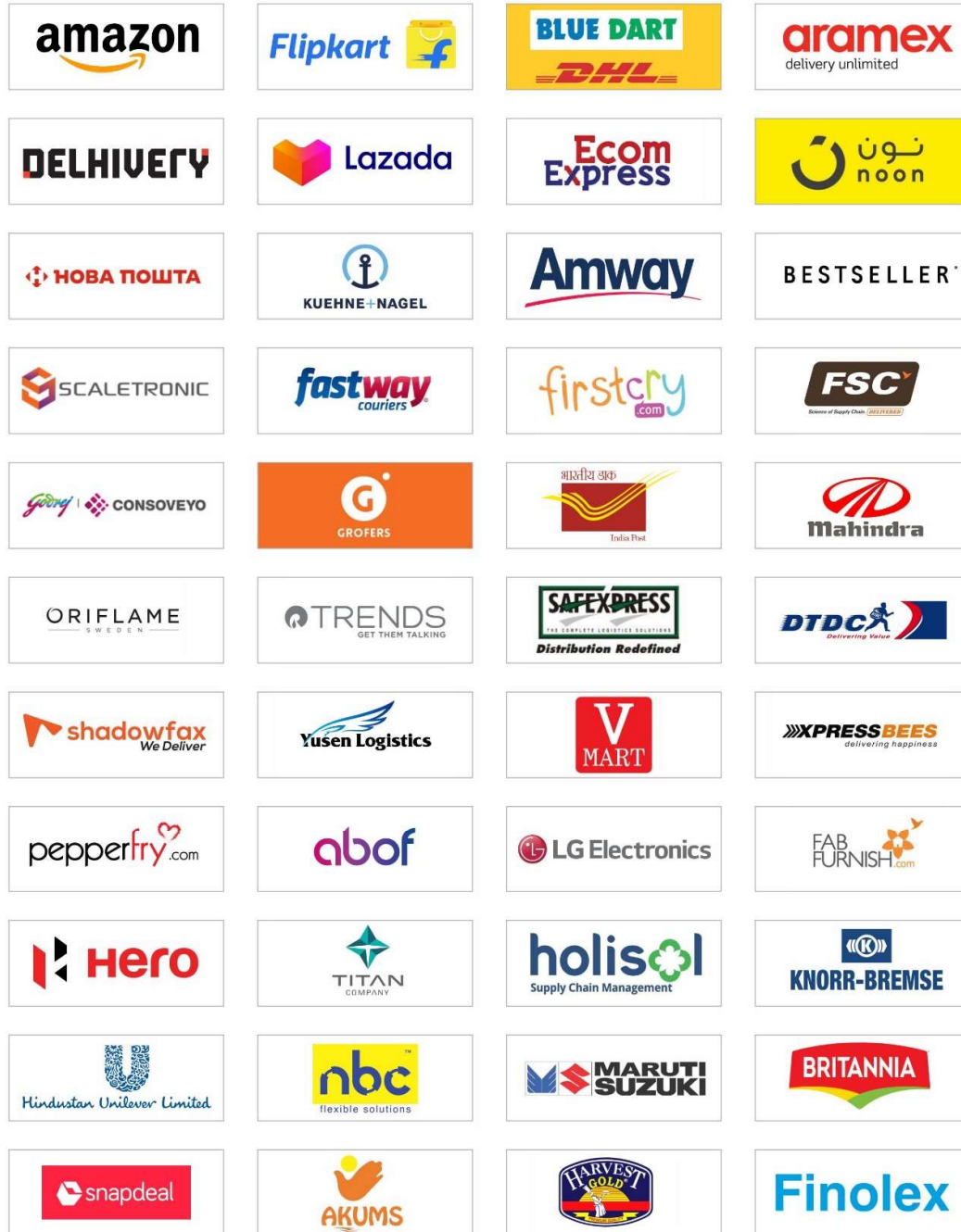
SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION

SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.



7. Handled Parcel Spectrum

7.1 Product size Conveyable on the entire system.

Max Length	mm	810
Max Width	mm	665
Max Height	mm	420
Max Weight	Kg	40
Min length	mm	210
Min Width	mm	145
Min Height	mm	75
Min Weight	Kg	2

7.2 Parcels to be loaded on Conveyor/Sorter shall have the following characteristics:

1. Centre of Gravity of items must not move during conveyance or sorting.
2. Item must not have magnetic content; otherwise behaviour of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic conveying/sorting system or damaging the system. They must be robust enough to avoid disintegration of container material and loose of contents in the conveying/sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

7.3 Parcel not loadable on the system

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

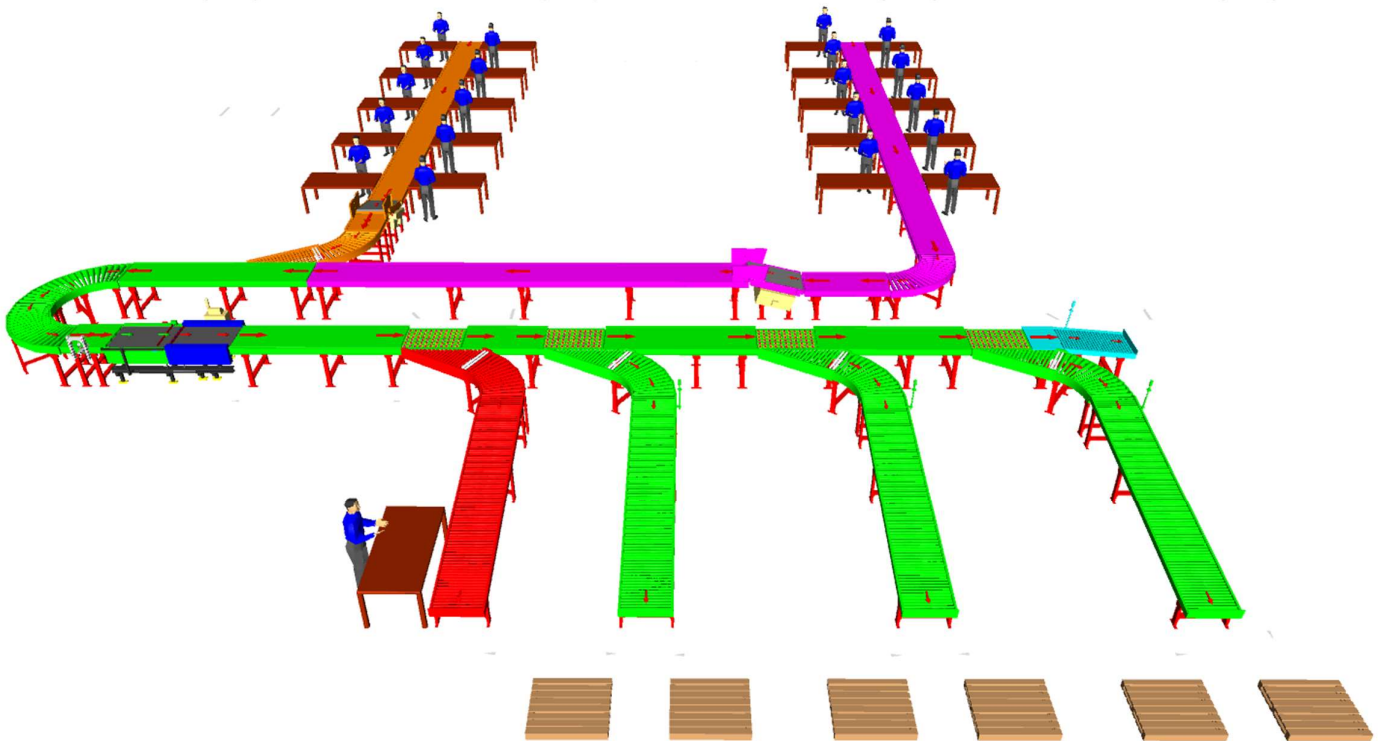
7.4 Types of Products

Parcel Type	Description
Cartons	Types of Carton Min: L-210 X W-145 X H-75 MM Max: L-810 X W-665 X H-420 MM

8. Proposed System Description

8.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing and acceptance testing of the complete automation system to handle Cartons as per specifications shared by Festo.



8.2 Summary of the System

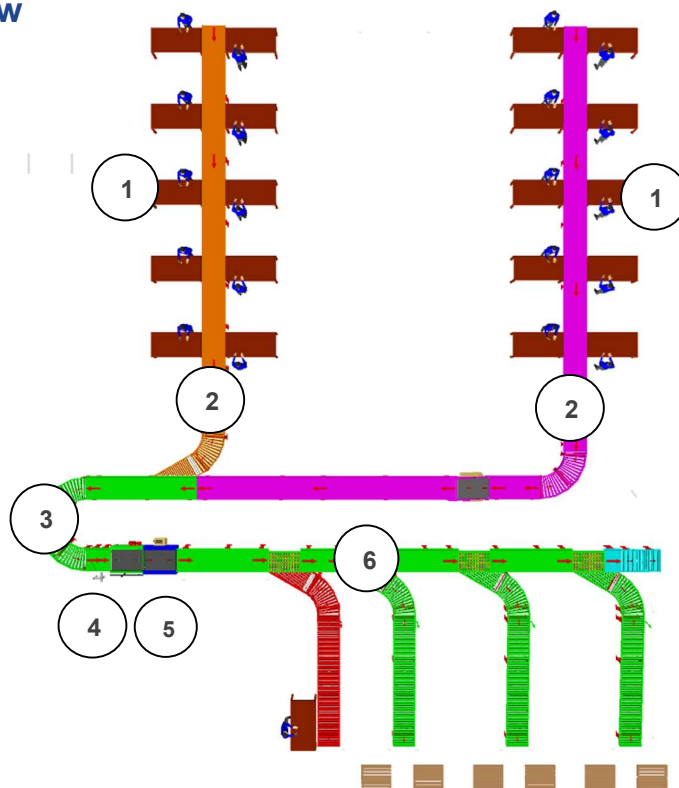
1. **Infeed:** SKU Cartons are fed into the system through 2 Nos Infeed lines running throughout the packing table area.
2. **Carton Weighing:** Cartons entering the main high undergoes dynamic weighing of each carton & the weigh is recorded in real time.
3. **Outbound Sorter:** Cartons are sorted Cross Dock and Beat wise through SWEDI Diverter into three main aisles.
4. Provision for rejection line for inspection & rework is given separately.
5. Distance b/w Scanner and Divert units is kept considering communication through PLC. (Stackbox to ensure WCS to have required data on regular basis)

8.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Maximized sorting capacity throughout at dispatch sorter to meet the requirement.
3. Low occupancy of floor space in the building
4. FALCON's system can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear

9. Layout View



Legend

- 1 – Packing Table
- 2 – Infeed loading line
- 3 – Highway line
- 4 – Side Barcode Scanner
- 5 – Weighing Conveyor
- 6 – Outbound Sorter

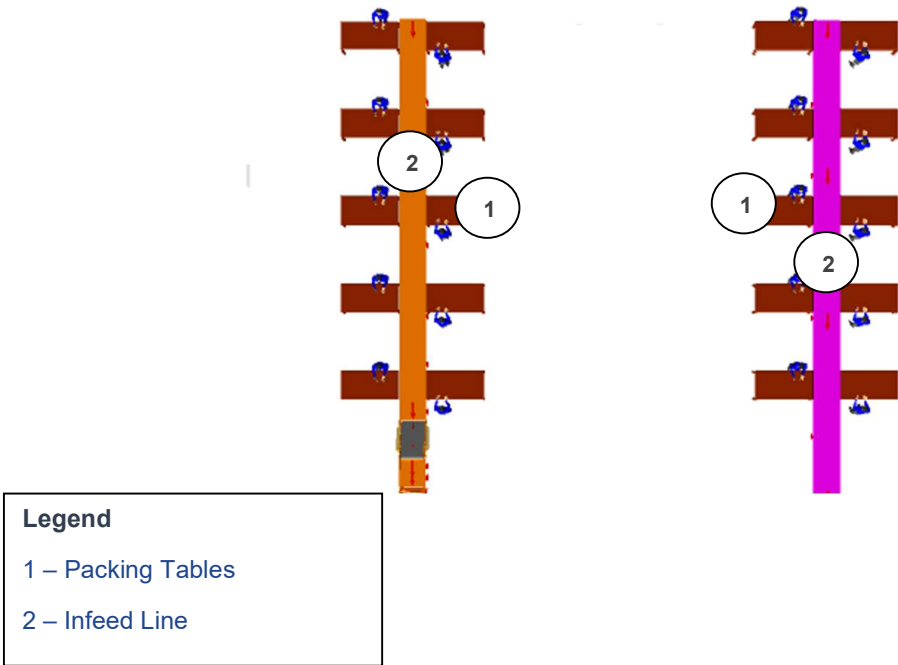
10. Proposed System Capacity

10.1 System Capacity at various areas

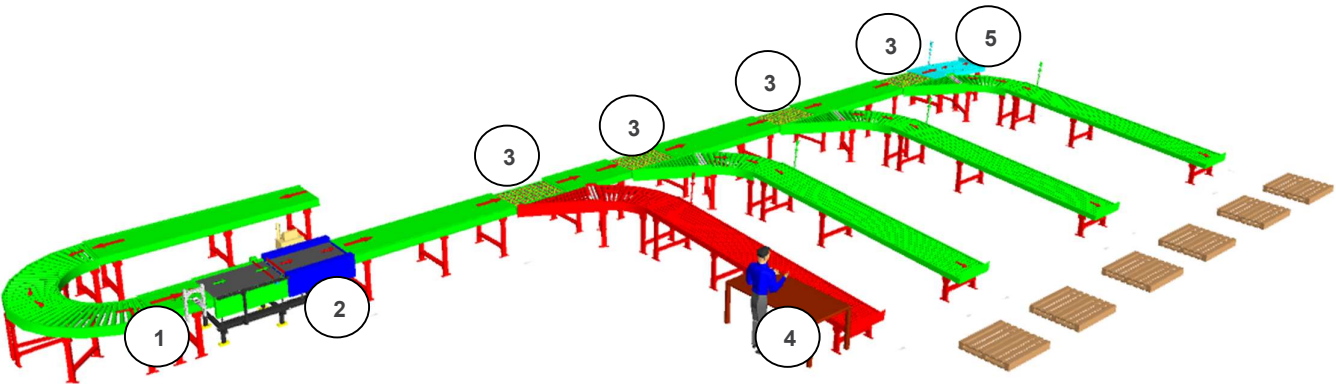
Activity	Cartons per hour
Infeed Lines and Loop	1000+1000
Weighing Conveyor	2000
Outbound Sorter	2000

11. System description

11.1 Infeed Lines and Loop



11.2 Outbound Sorter



Legend

1 – Highway Line

2 – Barcode Scanner

3 – Weighing Conveyor

4 – SWEDI Units

5 – Rework Line

6 – Sortfail Line

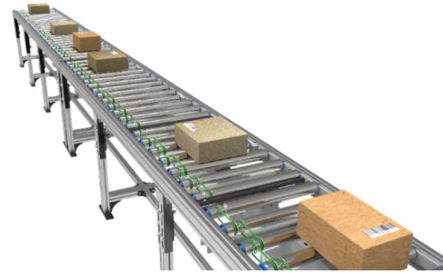
12. System Components

12.1 Powered Roller Conveyor

Falcon's Roller conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI
- Comes with MS frame.



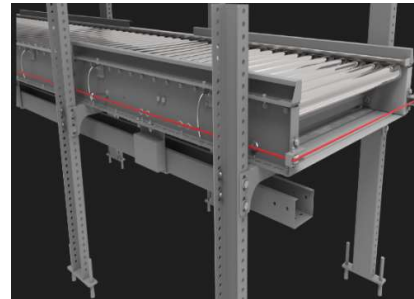
Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	800
Conveyor Height	mm	Variable as per design shared
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Poly V Belt Drive
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Load capacity/unit length of Conveyor	kg/m	50
Roller Span	mm	Variable
Roller Diameter	mm	50
Pitch of Rollers	mm	75/100
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions

12.2 Accumulation Conveyor

Falcon's Accumulation conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's accumulation conveyor:

- Low Noise
- Maximum uptime.
- High safety standards.
- Fastest ROI
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	Mm	800
Conveyor Height	Mm	As per Layout requirement
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Round/Poly V Belt Drive
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Load capacity per unit length of Conveyor	kg/m	50
Roller Span	Mm	Approx. 600
Roller Diameter	Mm	Approx. 50
Pitch of Rollers	Mm	75
Zone Length	Mm	825
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions
Type of Guides	Type	Steel guides at 50mm height basis the position of conveyor on both sides from Conveyor Belt Surface.

12.3 PVC Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	800
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Pitch on return side (Bottom)	mm	1200
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Tensioning Rollers	Spec	Dia 81.2 x 2.85 Thick, 25mm Shaft Dia, & Bearing Size-UCT205D1, MOC-MS, Zinc Plated -25 microns

Type of Drive	Type	Rubberized
Angle of wrap/Contact - Belt with Drive pulley	Deg	>180 deg
Motor Shaft	MOC	EN9 Shaft material,
Load capacity per unit length of Conveyor	kg/m	50
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradley
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

12.4 Powered Roller Turns & Merges



12.5 Side Barcode Scanners

System is integrated with automated inline side barcode scanning system capable for scanning 1D barcodes with high accuracy.



Specification	UOM	Remark
Manufacturer Name	Name	Cognex/Sick
Barcode Type	Type	1D
Location	Side	Left
Orientation	Type	Omnidirectional
Color of bars	Color	Black
Under Foil	yes/no	No

12.6 Weighing Scale

High Accuracy weighing scale for inline weight scanning.



Specification	UOM	Remark
Manufacturer Name	Name	Mettler Toledo/Bizerba
Minimum Weight	Kg	0.04
Maximum Weight	Kg	40

12.7 Swivel Wheel Sorting Unit – Falcon

Falcon's Swivel wheel divert is used in system for the smooth transfer of products.

Some Salient Features of Falcon's Swivel Wheel Divert:

- Low Noise
- Maximum uptime.
- Electric based rotation of wheels.
- Minimal maintenance
- High safety standards.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Type - Equipment	Type	Merge/Sorting
Type - Operation	Type	Sorting
Mechanism for Swivel	Type	Electric servo motor
Roller drive	kW	Powered smart wheel
Roller pitch (L x W)	Type	110 x 120
No. of divert sections	Yes/No	1 or 2
Max. product weight (kg/m2)	kg/m2	25
Throughput	PPH	2000
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

13. Proposed System Technical Details/BOM

13.1 Mechanical equipment

Pos.	Qty.	Description	Value
1	1	Infeed System - PVC Belt Conveyor - Accumulation Roller Conveyor 60 deg turns - Angle Merge	~1.2 metres (01 Modules) ~ 14.8 metres (02 Modules) 01 Nos 01 Nos
2	1	SWEDI Sorter - Powered Roller Conveyor - Gravity Roller Conveyor 60 deg turns 90 deg turns - Angle Merge - SWEDI Unit - Side Scanner - Weighing Scale - Spacing Conveyor	~17.7 meters (07 modules) ~22.7 meters (05 modules) 04 Nos 02 Nos 04 Nos 04 Nos 01 Nos 01 Nos 01 Nos
3	1	Future Expansion - PVC Belt Conveyor - Accumulation Roller Conveyor 90 deg turns	~1.2 metres (01 Modules) ~ 29 metres (03 Modules) 01 Nos

13.2 Electrical Equipment

Electricals			
1	1	Consists of Main Power Distribution panel Main Control Panel Network Switches Field Cabling	Included

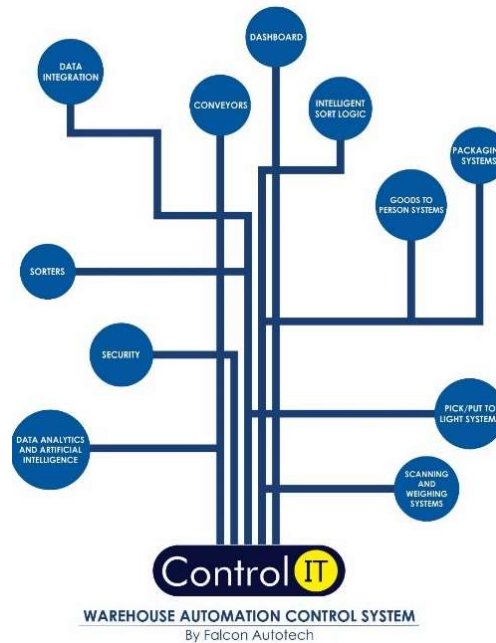
13.3 Control System

Components			
1	1	Consists of Siemen's PLC based Control System with Industrial Switch	1 Nos. As per requirement

14. High Availability Server Specifications

S. No	Configuration	Role	Qty	Purpose
1	Dell PowerEdge T440 (Chassis with up to 8, 3.5" Hot Plug Hard Drives) Tower Server Trusted Platform Module 1.2 Intel Xeon Silver 4110 2.1Ghz, 8C/16T, 9.6GT/s 2UPI, 11M Cache, Turbo, HT (85W) DDR4-2400 40 GB (2 x 16GB, 8 GB) RDIMM 2666MT/s Dual Rank 2 x 1TB SATA 7.2K 3.5" 6Gbps Fully assembled (Hot Plug) Hard Drive PERC H330 Integrated RAID Controller DVD +/-RW, SATA, Internal Single Hot-Plug Redundant Power Supplies (1+0) 495W Dual-Port 1GbE On-Board LOM iDRAC9 Express 3Yrs Basic Warranty - NBD	Node 1 - Prod Server	1	Failover Cluster is configured on this system. If this node fails, then the VM shifts from this node to another and vice versa
2	Dell PowerEdge T440 (Chassis with up to 8, 3.5" Hot Plug Hard Drives) Tower Server Trusted Platform Module 1.2 Intel Xeon Silver 4110 2.1Ghz, 8C/16T, 9.6GT/s 2UPI, 11M Cache, Turbo, HT (85W) DDR4-2400 40 GB (2 x 16GB, 8 GB) RDIMM 2666MT/s Dual Rank 2 x 1TB SATA 7.2K 3.5" 6Gbps Fully assembled (Hot Plug) Hard Drive PERC H330 Integrated RAID Controller DVD +/-RW, SATA, Internal Single Hot-Plug Redundant Power Supplies (1+0) 495W Dual-Port 1GbE On-Board LOM iDRAC9 Express 3Yrs Basic Warranty - NBD	Node 2 - Prod Server	1	Failover Cluster is configured on this system. If this node fails, then the VM shifts from this node to another and vice versa
3	Intel® Xeon® processor E3-1200 v5 product family, 16 GB RAM, 1 TB * 2HDD, RAID 1, iDRAC8 Enterprise (upgrade), 3 years Onsite warranty.	Domain Server	1	Hyper V will work with Domain System So that we required Domain Server Which will manage the nodes
4	Synology storage with 4 TB HDD	Network Storage Box 1	1	For Creation of Clustered Share Volume, VM will be created on this BOX
5	Synology storage with 4 TB HDD	Network Storage Box 2	1	Backup & Replication of CSV
6	Power Backup	Power Backup	3	UPS for the Servers & Network Boxes
7	Monitor - 19.5 inch	Monitor	1	Monitor
8	KVM Switch	Switch	1	To switch over the display between servers
9	Windows Server 2016 Standard R2 or Higher	Windows License	3	OS

15. Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

16. Key Component Makes

Component	Make
PLC & RIO	Siemens/Omron
VFD	Siemens/Omron/Lenze/AB
Ethernet switch	Siemens/Phoenix Contact
SMPS	Omron/Selec/Meanwell
Line chokes	EMIS/GURU NANAK ELE.
Slim relays	Phoenix contact
Terminals	Wago
Phase failure relay	Selec
Tower lamp + Hooter	Leuze
Utility switch & socket	Anchor
Fan/Air conditioner	Rexnord
Switchgear	Schneider
Main load break switch (CP)	Schneider
Geared Motors	SEW
MDR	Pulse
Cameras	Cognex/Sick
Weighing Scale	Mettler Toledo/Bizerba

17. Safety

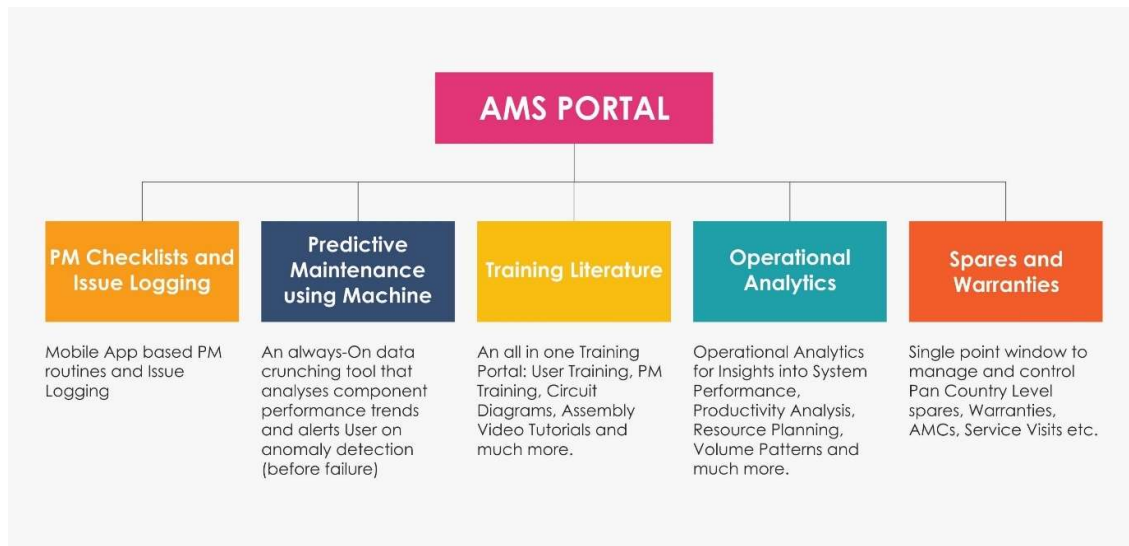
17.1 Push Buttons

Emergency Push Buttons will be provided at all the points of human interaction on the system.

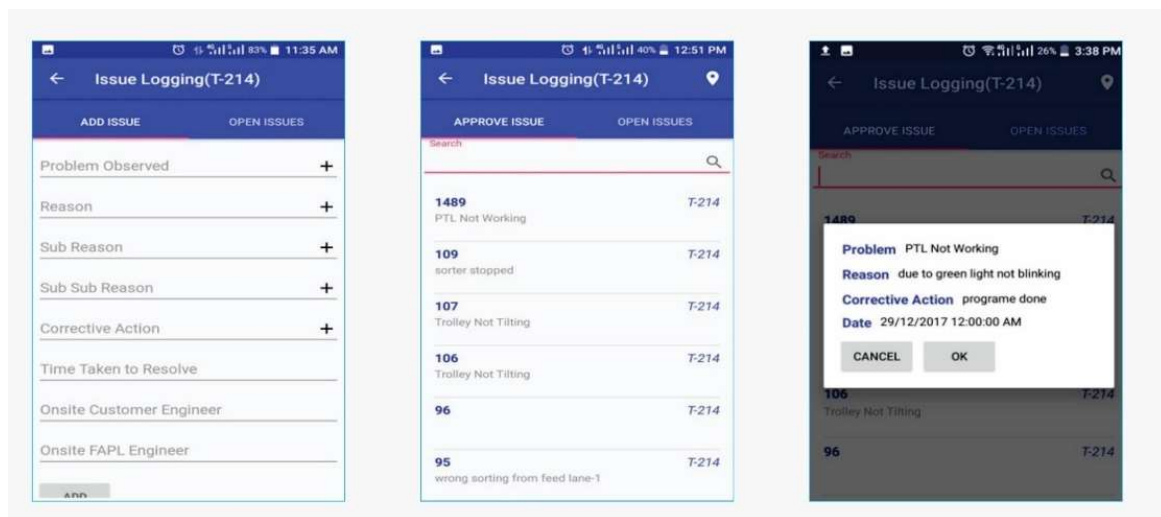


Safety Netting, barricading, Bollards, toe guards or any such safety requirements shall be in Festo Scope. Any other safety requirements requested by Festo will be on chargeable basis.

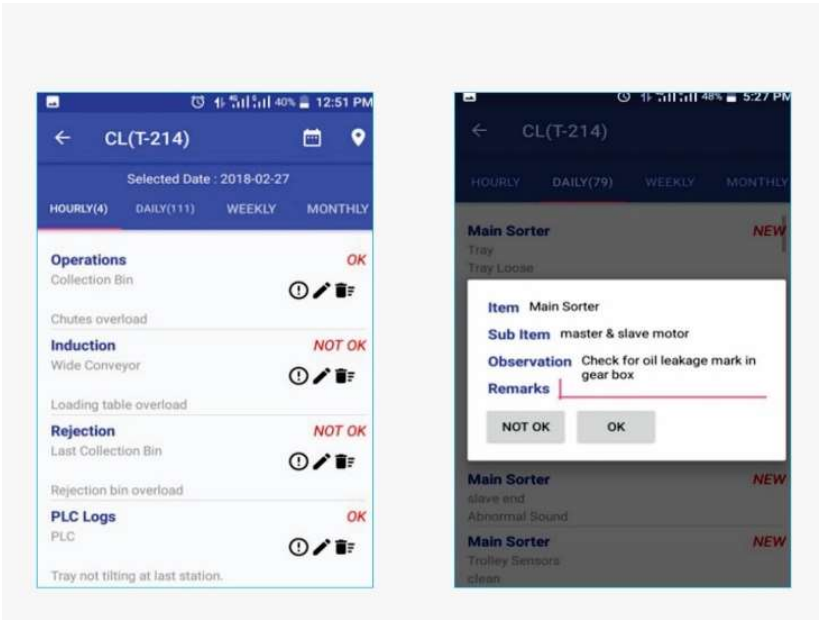
18. Automation Management System (AMS)



18.1 Issue Logging



18.2 Preventive Maintenance Check List



18.3 Predictive Maintenance



18.4 Training Portal

Dashboard

Home - Dashboard

Employee Training Admin

Employee Name: Training Type: Search

Show All Scheduled Completed

Employee Name	Training Type	Trainer	Notes	Training Taken By	Actual Start Date	Actual End Date	Status
Block 1001-1010	Product Training	TR Top Series TR Top Loop	Block 1001-1010	Web Chordays A			
Block 1001-1010	Product Training	System PTL Conveyor Belt Conveyor	Block 1001-1010	Web Chordays A			
Block 1001-1010	Product Training	BELT CONVEYOR WITH REJECTION AND JAM	Block 1001-1010	Web Chordays A			
Block 1001-1010	Product Training	CONVEYOR SAG SORTER SYSTEM	Block 1001-1010	Web Chordays A			
Block 1001-1010	Product Training	TR Top Series SIEP SORTER SAG SORTER SYSTEM PTL POPUP SORTER PTL	Block 1001-1010	Web Chordays A			
Block 1001-1010	Product Training	VUL Custom Conveyor Check	Block 1001-1010	Web Chordays A			
Block 1001-1010	Product Training	Wagon Belt Type Check Diagram	Block 1001-1010	Web Chordays A			
Block 1001-1010	Product Training	Carton Belt Conveyor	Block 1001-1010	Web Chordays A			
Block 1001-1010	Product Training	TR Top Series SIEP SORTER SAG SORTER SYSTEM	Block 1001-1010	Web Chordays A			

Dashboard

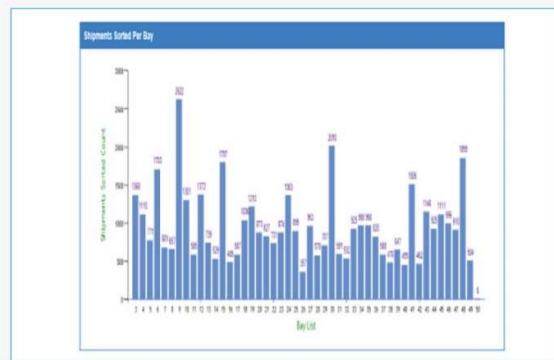
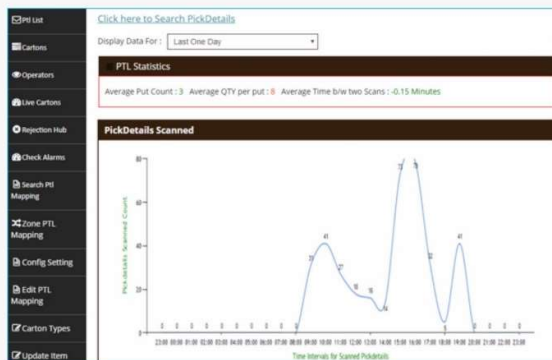
Home - Dashboard

Training Material Admin

Training Type: Document Name: Price Name: Search

Training Type	Subject	Document Name	Training Document	Video Name	Training Video	Time Limit	Setting
Component Training	KIR TRAINING	SPACED MANUAL	download			2	100%
Component Training	SPC TRAINING	POWERED AIR MANUAL	download			2	100%
Component Training	PTL	PTL INSTRUCTION MANUAL	download			2	100%
Component Training	PTL	operational manual	download			2	100%
Product Training	TR Top Series	1-214 SAG SORTER setting	download			2	100%
Maintenance & Troubleshooting	TR Top Series	1-214 SAG SORTER MANUAL	download			2	100%
Product Training	TR Top Series	1-214 SAG SORTER MANUAL	download			2	100%
Product Training	TR Top Series	1-214 REDUCTION LINE MANUAL	download			2	100%
Department Procedures	Field & Panel Wiring	Electrical 01	download			20 mins	100%

18.5 Operational Analysis



19. Program Organisation

Note- Approx. 60-90 days required to handover the system from the date of formal confirmation.

19.1 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with FESTO
3. Scheduling of the resource management.
4. Management of the risks and opportunities.
5. Management of the list of anomalies or reservations.

20. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use) and post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

21. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Servers
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms- If not specified
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.